

Choice of Anesthesia for Cesarean Delivery: An Analysis of the National Anesthesia Clinical Outcomes Registry

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Neuraxial anesthesia use in cesarean deliveries (CDs) has been rising since the 1980s, whereas general anesthesia (GA) use has been declining. In this brief report we analyzed recent obstetric anesthesia practice patterns using National Anesthesiology Clinical Outcomes Registry data. Approximately 218,285 CD cases were identified between 2010 and 2015. GA was used in 5.8% of all CDs and 14.6% of emergent CDs. Higher rates of GA use were observed in CDs performed in university hospitals, after hours and on weekends, and on patients who were American Society of Anesthesiologists class III or higher and 18 years of age or younger. (Anesth Analg 2017;124:1914–7)

Historically, the type of anesthesia used during cesarean deliveries (CDs) has been an important determinant of maternal and fetal outcomes.¹ In the United States, general anesthesia (GA) use has been steadily declining since the 1980s and is now largely supplanted by neuraxial anesthesia (NA).^{2–4} Single-institution and New York state data have shown continued declines in GA rates within at least the first decade of the 21st century.^{5,6} The recently published 30-year obstetric anesthesia workforce survey, inclusive of data up to 2011, showed a stably low rate of elective GA but an apparent rise in emergent GA use from 15% to 19% during the preceding decade.⁴

The National Anesthesiology Clinical Outcomes Registry (NACOR), a national repository for deidentified administrative, clinical, and quality-focused data formed in 2010 from anesthesia practices in the United States, allows easy electronic access to national anesthesia practice information at a lower cost and less effort than traditional methods for acquiring a similar breadth of data. Our goal in this brief report is to review anesthesia practice data for CD cases submitted to NACOR and to evaluate the feasibility of identifying facility and patient factors that may influence anesthesia choice.

METHODS

Data Source

This research was approved by our institutional review board and was exempted from the consent requirement. We analyzed records of CDs submitted to NACOR with current procedural terminology (CPT) codes 59510, 59514, 59515,

59618, 59620, or 59622 that occurred between January 1, 2010, and March 31, 2015. All CD cases were stratified by American Society of Anesthesiologists (ASA) physical status (PS) and the presence of E status, denoting emergency surgery. We note that ASA effectively eliminated the ASA I class for obstetric anesthesia in 2014, but the majority of the NACOR data predated that policy change.⁷ Additional case elements analyzed separately by all CDs included patient age, trial of labor after CD (TOLAC; CPT codes 59618, 59620, and 59622), year of surgery, time of surgery (7:00 AM to 5:00 PM vs 5:01 PM to 6:59 AM) as defined by anesthesia start time, day of week (weekend versus weekday), US region (Northeast, Midwest, South, or West), facility volume (1000 or fewer CDs versus >1000 CDs annually), setting (urban versus rural), and facility type. The facility type options included university hospitals, large community hospitals (consisting of >500 beds), medium community hospitals (100 to 500 beds), and other facility types, inclusive of small community hospitals (<100 beds) and specialty hospitals.

Statistical Analysis

R Project for Statistical Computing (R version 3.1.2) was used to perform all statistical analyses. We fitted multiple univariable logistic regression models to test whether each case characteristic, represented categorically, was associated with anesthesia type for all CDs. Results from the logistic regression were reported as odds ratios (ORs) with 95% confidence intervals (CIs). All characteristics that had a statistically significant association were used as covariates to fit one multivariable logistic regression model. A significance level of $P < .0001$ was used to avoid false positive findings in a multivariable model. Missing ASA PS values were imputed to a value of 2, and missing age was imputed to mean age of dataset (equal to 30 years). Missing values in nonordered categorical variables were coded as unknown. The area under the receiving operating characteristic curve (AUC) was calculated to estimate the performance of the regression model.

RESULTS

NACOR provided 287,127 CDs out of 26,568,734 total cases from January 2010 to March 2015.^{8,9} After exclusion of CD cases with unknown primary anesthesia type, the sample

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size was 218,285, with 205,671 (94.2%) cases performed under NA and 12,614 (5.8%) cases performed under GA. A total of 308 different hospital facilities contributed cases to this cohort, with a range from 1 to 14,385 reported cases per facility. A total of 15,282 emergent CD cases were identified, of which 13,046 (85.4%) cases were performed under NA and 2236 cases (14.6%) performed under GA. Temporal and total use of GA is depicted in the Figure.

Distribution of anesthesia rates (GA versus NA) by case characteristics for all CD cases is shown in Table 1. A summary of univariable and multivariable logistic regression analyses for all CD cases is shown in Table 2. The AUC for the logistic regression model was 0.645 (95% CI, 0.640–0.650).

DISCUSSION

The overall rate of GA usage nationwide comprised a small fraction (5.8%) of total anesthesia procedures performed during CDs. The GA rate at university hospitals was highest (8.5%) among facility types, an observation that was surprising because the GA usage rates spanning from 2000 to 2005 published previously based on a single university hospital ranged from 0.4% to 1%.⁶ The rate of GA usage during emergent CD reported in the NACOR dataset (14.5%) is consistent with more recently published work by the obstetric anesthesia work force survey.⁴ Interestingly, the 30-year update from the workforce survey reported a slight increase in emergent GA use from 15% to 19%; however, the statistical significance of this rate increase is not discussed in that article.

Although NACOR is expected to capture at least basic data from approximately 25% of all anesthesia cases in the United States in 2015,¹⁰ the number of CD cases included over the period of January 2010 to March 2015 (287,127) appears to represent a small proportion of the estimated number of CDs performed in the United States during the same period (6.8 million).¹¹ The annual number of CD cases in NACOR more than doubled between 2010 and 2014, indicating evolving reporting practices. These observations suggest that although the NACOR data allow for the identification of subsets of data elements that are associated with higher rates of GA use, our statistical analysis can only be considered illustrative; results may change as the sample

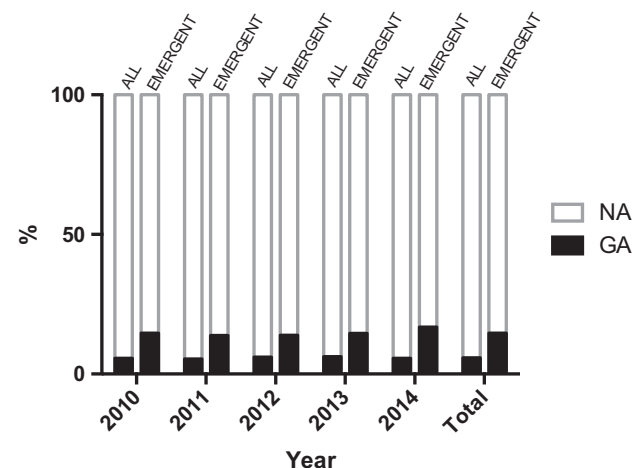


Figure. Percentage of all cesarean deliveries and emergent cesarean deliveries performed under general anesthesia versus those performed under neuraxial anesthesia by year.

Table 1. Patient and Facility Characteristics Presented by NA or GA Selection for All CDs

All CDs	NA		GA	
	N	%	N	%
Total	205,671	94.22	12,614	5.78
Age				
Mean age (± SD)	30.02 ± 5.95		29.26 ± 6.30	
≤18 years	3319	1.61	366	2.90
19–34 years	154,092	74.92	9502	75.33
≥35 years	48,260	23.46	2746	21.77
ASA PS class				
ASA I–II	176,836	85.98	8983	71.21
ASA III–V	28,835	14.02	3631	28.79
Time of day				
Day shift (7:00 AM–5:00 PM)	141,901	68.99	7177	56.90
After hours (5:01 PM–6:59 AM)	63,770	31.01	5437	43.10
Time of week				
Weekday	176,812	85.97	10,230	81.10
Weekend	28,859	14.03	2384	18.90
Urban versus rural				
Urban	73,948	35.95	4229	33.53
Rural	114,278	55.56	7030	55.73
Unknown	17,445	8.48	1355	10.74
Facility type				
University hospital	18,088	8.79	1685	13.36
Large community	45,204	21.98	2196	17.41
Medium-sized community	81,514	39.63	5543	43.94
Other facility	46,980	22.84	2690	21.33
Unknown facility	13,885	6.75	500	3.96
United States region				
Northeast	42,729	20.78	2515	19.94
Midwest	40,729	19.80	2628	20.83
South	81,869	39.81	5802	46.00
West	26,406	12.84	1169	9.27
Unknown region	13,938	6.78	500	3.96
CD volume per year				
Volume ≤1000	149,312	72.60	9301	73.74
Volume >1000	56,359	27.40	3313	26.26
Emergency status				
Not emergency	192,625	93.66	10,378	82.27
Emergency	13,046	6.34	2236	17.73
TOLAC				
Not TOLAC	204,342	99.35	12,588	99.79
TOLAC	1329	0.65	26	0.21
Year of CD				
2010	26,397	12.83	1668	13.22
2011	32,728	15.91	1965	15.58
2012	44,516	21.64	2912	23.09
2013	40,028	19.46	2618	20.75
2014	55,162	26.82	3127	24.79
2015	6840	3.33	324	2.57

Abbreviations: ASA PS, American Society of Anesthesiologists Physical Status; CD, cesarean deliveries; GA, general anesthesia; NA, neuraxial anesthesia; TOLAC, Trial of Labor After Cesarean Delivery.

collected by NACOR expands. One observation supporting the validity of the NACOR dataset is that the derived emergent GA rate is similar to obstetric anesthesia work force survey results from the preceding decade.⁴

A few additional methodologic limitations warrant discussion. NACOR does not allow us to identify subsets of GA cases resulting from NA failure, which can vary widely between institutions (2% to 20%).^{12–14} NA placed for vaginal deliveries that were subsequently converted to CDs

Table 2. Results of a Univariate and Multivariate Logistic Regression Model Determining Association of Patient and Facility Factors With the Use of NA Versus GA as Primary Anesthetic for All CDs

All Cesarean Deliveries	Univariable Analysis		Multivariable Analysis	
	OR (95% CI)	P Value	OR (95% CI)	P Value
Age				
≤18 years old	1.79 (1.60–2.00)	<.0001	1.62 (1.45–1.82)	<.0001
19–34 years old	Reference		Reference	
≥35 years old	0.92 (0.88–0.96)	.0003	0.91 (0.87–0.95)	<.0001
ASA PS class				
ASA I–II	Reference		Reference	
ASA III–V	2.48 (2.38–2.58)	<.0001	2.34 (2.24–2.43)	<.0001
Time of day				
Day shift (07:00–17:00)	Reference		Reference	
After hours (17:01–06:59)	1.69 (1.63–1.75)	<.0001	1.42 (1.37–1.48)	<.0001
Day of week				
Weekday	Reference		Reference	
Weekend	1.43 (1.36–1.50)	<.0001	1.24 (1.19–1.31)	<.0001
Urban versus rural				
Urban	Reference		–	–
Rural	1.08 (1.03–1.12)	.0003	–	–
Unknown	1.36 (1.27–1.45)	<.0001	–	–
Facility type				
University hospital	Reference		Reference	
Large community	0.52 (0.49–0.56)	<.0001	0.49 (0.46–0.53)	<.0001
Medium-sized community	0.73 (0.69–0.77)	<.0001	0.76 (0.72–0.81)	<.0001
Other facility	0.61 (0.58–0.65)	<.0001	0.67 (0.63–0.72)	<.0001
Unknown facility	0.39 (0.35–0.43)	<.0001	0 (0 to ∞)	.85
US region				
Northeast	Reference		Reference	
Midwest	1.10 (1.04–1.16)	.0014	1.25 (1.17–1.33)	<.0001
South	1.20 (1.15–1.26)	<.0001	1.34 (1.27–1.41)	<.0001
West	0.75 (0.70–0.81)	<.0001	0.77 (0.71–0.83)	<.0001
Unknown region	0.61 (0.55–0.67)	<.0001	0 (0 to ∞)	.84
CD volume per year				
Volume ≤1000	Reference		–	–
Volume >1000	0.94 (0.91–0.98)	.0054	–	–
Emergency status				
Not emergency	Reference		Reference	
Emergency	3.18 (3.03–3.34)	<.0001	2.96 (2.81–3.12)	<.0001
TOLAC				
Not TOLAC	Reference		Reference	
TOLAC	0.32 (0.22–0.47)	<.0001	0.31 (0.21–0.46)	<.0001
Year of CD				
2010	Reference		Reference	
2011	0.95 (0.88–1.02)	.1360	0.92 (0.86–0.99)	.02
2012	1.04 (0.97–1.10)	.2700	1.00 (0.94–1.06)	.96
2013	1.04 (0.97–1.10)	.2900	0.99 (0.93–1.06)	.82
2014	0.90 (0.84–0.95)	.0005	0.88 (0.82–0.94)	<.0001
2015	0.75 (0.66–0.85)	<.0001	0.74 (0.65–0.84)	<.0001

ORs with associated 95% CIs are provided for each category with associated reference variable, noted under the Reference column. An OR with 95% CI not including 1.00 is considered statistically significant. P values are a result of χ^2 analysis for categorical variables, and Student t test for continuous variables. P < .0001 is considered statistically significant.

Abbreviations: ASA PS, American Society of Anesthesiologists Physical Status; CD, cesarean delivery; CI, confidence interval; OR, odds ratio; TOLAC, trial of labor after cesarean delivery; US, United States.

also cannot be distinguished from NA placed de novo for CDs. Consequently, the relationship between institutional characteristics and use of GA is likely confounded by the unmeasured relative contributions of elective CD and unplanned CD in each institution. For CD-type designations, NACOR also does not explicitly enforce a standard definition for the emergent classification. The American Congress of Obstetricians and Gynecologists and American Academy of Pediatrics both define emergent as the interval from decision to incision of <30 minutes; however, minor

variations to the 30-minute rule have been reported across practices. We did observe from the NACOR dataset that emergent status increases the likelihood of GA use when TOLAC decreases the likelihood of GA use.

In summary, NACOR is the largest national database for anesthesia practice information and the best available data source to help identify current practice patterns, outcomes, and eventually longitudinal practice trends. Future studies will be needed to validate our observations against independent national data sources. ■■

DISCLOSURES

Name: Jeremy Juang, MD, PhD.

Contribution: This author helped design and conduct the study, and helped with data collection and analysis, and manuscript preparation.

Conflicts of Interest: None.

Name: Rodney A. Gabriel, MD.

Contribution: This author helped design and conduct the study, and helped with data collection, analysis, and manuscript preparation.

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Name: Richard P. Dutton, MD, MBA.

Contribution: This author helped with data analysis and manuscript preparation.

Conflicts of Interest: Richard P. Dutton is a member of the Data Use Committee, Anesthesia Quality Institute.

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Contribution: This author helped with data analysis and manuscript preparation.

Conflicts of Interest: None.

Name: Richard D. Urman, MD, MBA.

Contribution: This author helped design and conduct the study, and helped with data collection, analysis, and manuscript preparation.

Conflicts of Interest: Richard D. Urman is a member of the Data Use Committee, Anesthesia Quality Institute.

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